sound sculpture

ENGN0620 - Design Brief

TEAM

Kareem El Naggar
Emelie Nguyen
Alexia Bodet
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MENTORS

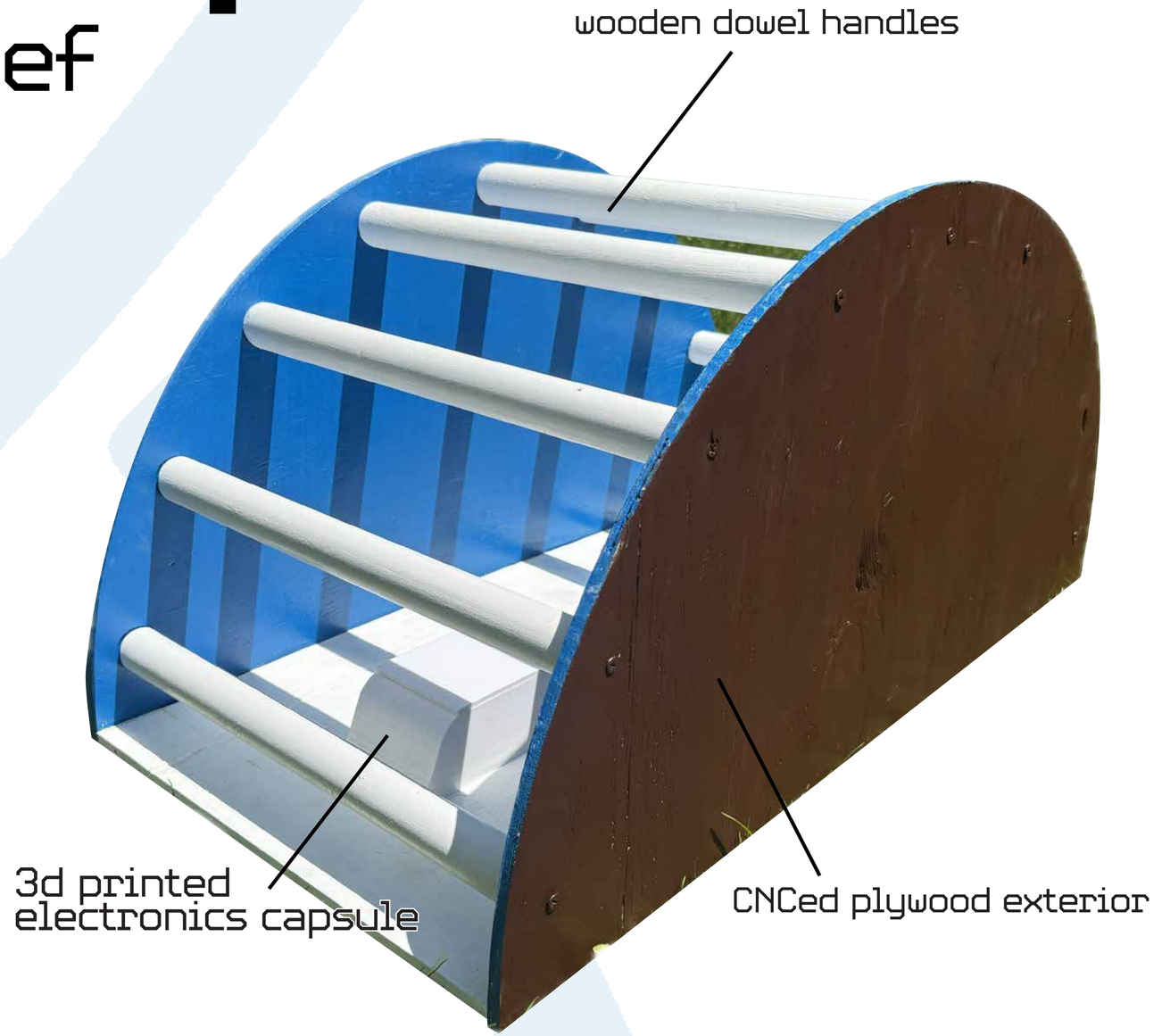
Maia Schlussel (TA)
Dr. Louise Manfredi
Dr. Daniel Harris

TIMELINE

March 2026 - May 2026

SKILLS

rapid prototyping, embedded programming, electronics, woodworking, user research



overview

client: Professor Ja-Nae Duane

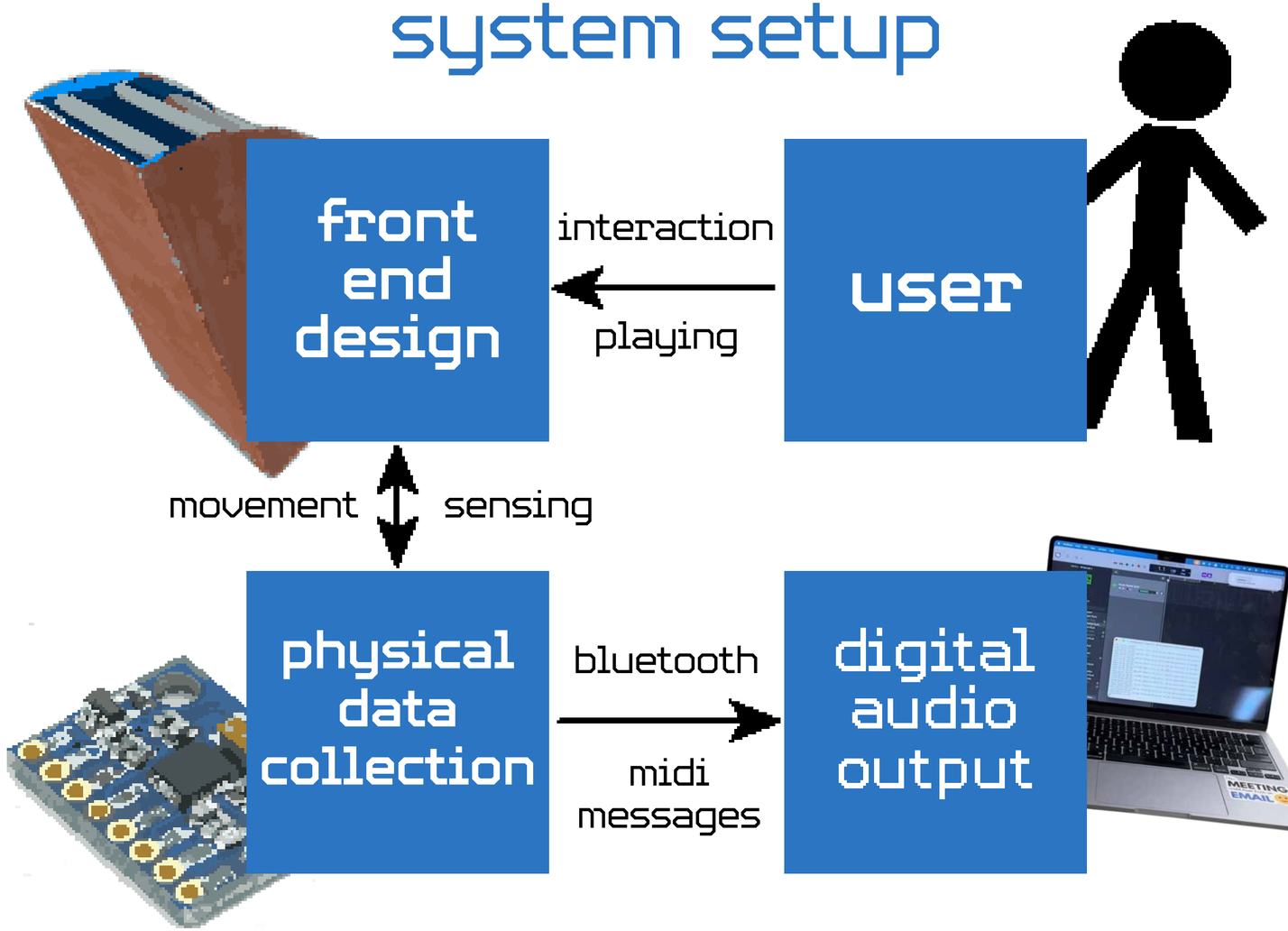
project brief

create an interactive installation that combines visual, auditory, and tactile

my role

- led physical prototyping
- developed midi interfacing mechanism for audio
- soldered and assembled electronics
- iterated front-end design

system setup

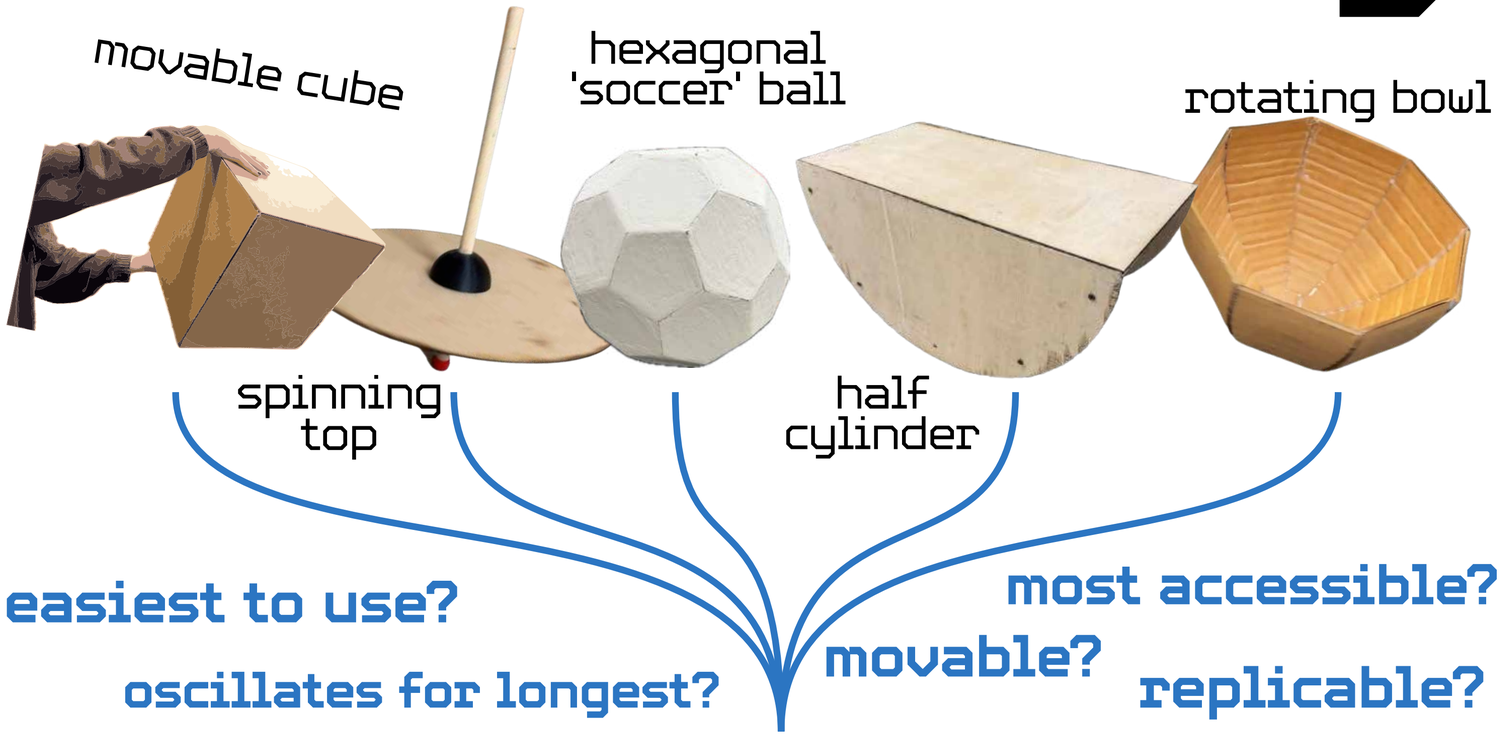


iteration & testing

We developed gyroscope-based midi technology for this project to translate rotational data into music.

Thus, our physical prototyping methodology revolved (no pun intended) around objects that spin, rotate, and/or pivot to activate sound.

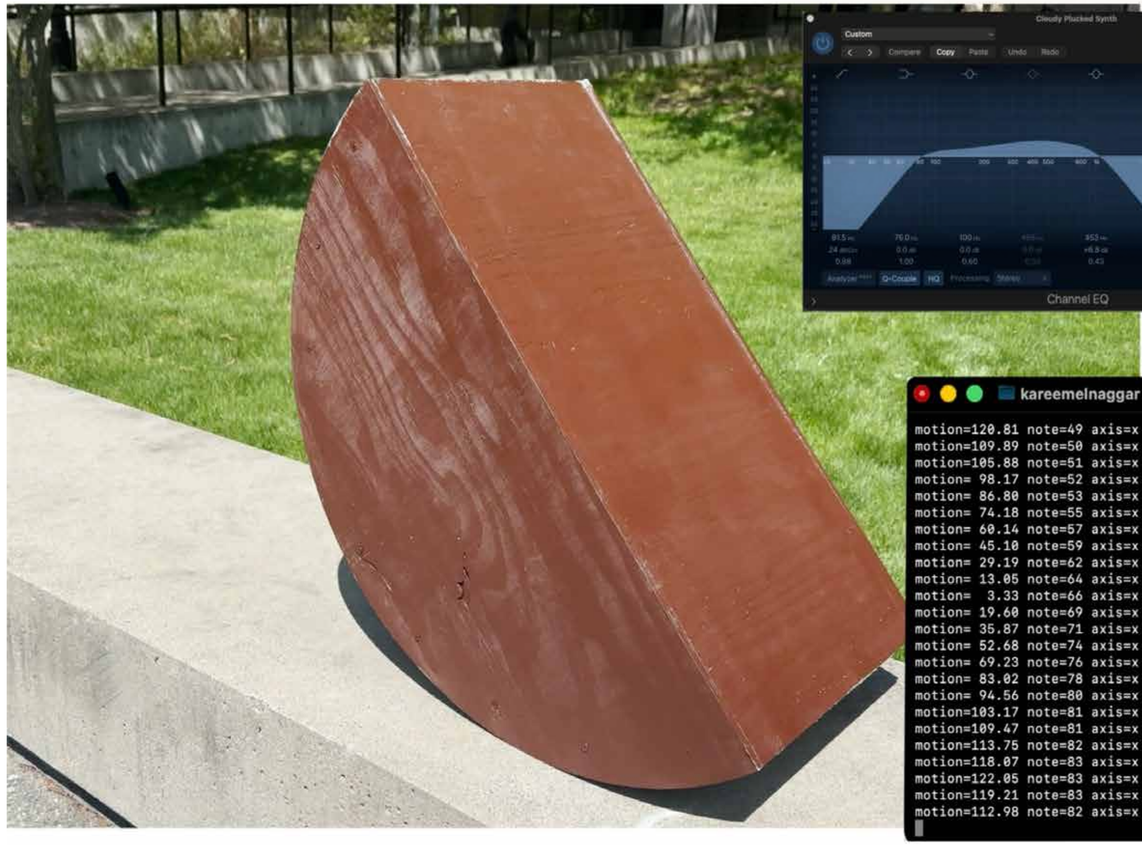
Researching different ways to fabricate rolling/curved objects and learning to build them using various techniques, along with user-testing user interaction for the most immersive experience, we landed on a final design.





rocking half cylinder shape

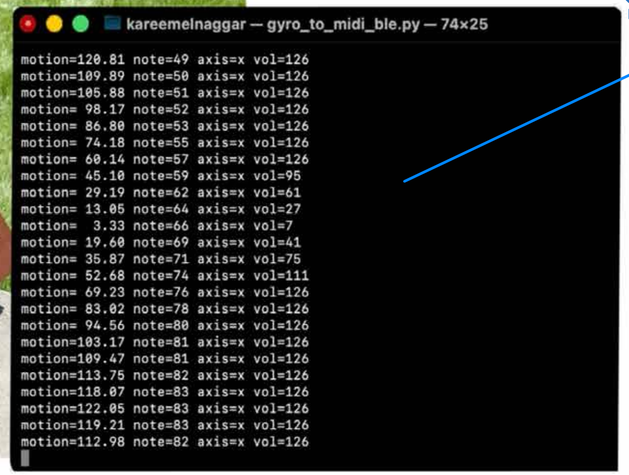
results





audio output

gyroscope data



moving forward, to complete the installation we will build more of them, each painted different colors, and allow them to each result in unique musical outputs to provide variety